[36] J. Mao, H. Jonathan, A. Toshev, O. Camburu, A. Yuille, and K. Murphy. Generation and comprehension of unambiguous object descriptions. In *CVPR*, 2016. 2

[37] Matterport. Press release, October 2017. 8

[38] H. Mei, M. Bansal, and M. R. Walter. Listen, attend, and walk: Neural mapping of navigational instructions to action sequences. In *AAAI*, 2016. 1, 2

[39] A. H. Miller, W. Feng, A. Fisch, J. Lu, D. Batra, A. Bordes, D. Parikh, and J. Weston. Parlai: A dialog research software platform. *arXiv preprint arXiv:1705.06476*, 2017. 4

[40] P. Mirowski, R. Pascanu, F. Viola, H. Soyer, A. Ballard, A. Banino, M. Denil, R. Goroshin, L. Sifre, K. Kavukcuoglu, et al. Learning to navigate in complex environments. In *ICLR*, 2017. 6

[41] D. K. Misra, J. Langford, and Y. Artzi. Mapping instructions and visual observations to actions with reinforcement learning. In *EMNLP*, 2017. 1, 3, 6

[42] P. K. Nathan Silberman, Derek Hoiem and R. Fergus. Indoor segmentation and support inference from rgbd images. In *ECCV*, 2012. 3

[43] D. A. Norman. *The Design of Everyday Things*. Basic Books, Inc., New York, NY, USA, 2002. 5

[44] S. J. Rennie, E. Marcheret, Y. Mroueh, J. Ross, and V. Goel. Self-critical sequence training for image captioning. In *CVPR*, 2017. 7

[45] S. Ross, G. Gordon, and D. Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *AISTATS*, 2011. 7

[46] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei. Imagenet large scale visual recognition challenge. *IJCV*, 2015. 6

[47] M. Savva, A. X. Chang, A. Dosovitskiy, T. Funkhouser, and V. Koltun. MINOS: Multimodal indoor simulator for navigation in complex environments. *arXiv:1712.03931*, 2017. 3, 4

[48] S. Song, S. P. Lichtenberg, and J. Xiao. SUN RGB-D: A rgb-d scene understanding benchmark suite. In *CVPR*, 2015. 3

[49] I. Sutskever, O. Vinyals, and Q. V. Le. Sequence to sequence learning with neural networks. In *NIPS*, 2014. 6

[50] L. Tai and M. Liu. Towards cognitive exploration through deep reinforcement learning for mobile robots. *arXiv preprint arXiv:1610.01733*, 2016. 3

[51] M. Tapaswi, Y. Zhu, R. Stiefelhagen, A. Torralba, R. Urtasun, and S. Fidler. MovieQA: Understanding stories in movies through question-answering. In *CVPR*, 2016. 2

[52] S. Tellex, T. Kollar, S. Dickerson, M. R. Walter, A. G. Banerjee, S. J. Teller, and N. Roy. Understanding natural language commands for robotic navigation and mobile manipulation. In *AAAI*, 2011. 1, 2

[53] C. Tessler, S. Givony, T. Zahavy, D. J. Mankowitz, and S. Mannor. A deep hierarchical approach to lifelong learning in minecraft. In *AAAI*, pages 1553–1561, 2017. 3

[54] A. Torralba and A. A. Efros. Unbiased look at dataset bias. In *CVPR*, 2011. 4

[55] A. Vogel and D. Jurafsky. Learning to follow navigational directions. In *ACL*, 2010. 2

[56] D. Wierstra, A. Foerster, J. Peters, and J. Schmidhuber. Solving deep memory pomdps with recurrent policy gradients. In *International Conference on Artificial Neural Networks*, 2007. 6

[57] T. Winograd. Procedures as a representation for data in a computer program for understanding natural language. Technical report, Massachusetts Institute of Technology, 1971. 2

[58] Y. Wu, Y. Wu, G. Gkioxari, and Y. Tian. Building generalizable agents with a realistic and rich 3d environment. *arXiv:1801.02209*, 2018. 3

[59] C. Yan, D. Misra, A. Bennnett, A. Walsman, Y. Bisk, and Y. Artzi. CHALET: Cornell house agent learning environment. *arXiv:1801.07357*, 2018. 3

[60] Z. Yang, X. He, J. Gao, L. Deng, and A. J. Smola. Stacked attention networks for image question answering. In *CVPR*, 2016. 7

[61] A. R. Zamir, F. Xia, J. He, S. Sax, J. Malik, and S. Savarese. Gibson Env: Real-world perception for embodied agents. In *CVPR*, 2018. 3

[62] Y. Zhu, R. Mottaghi, E. Kolve, J. J. Lim, A. Gupta, L. Fei-Fei, and A. Farhadi. Target-driven visual navigation in indoor scenes using deep reinforcement learning. In *ICRA*, 2017. 1, 2, 3, 4, 6

Figure 9. Snapshot of the visual diversity in the Matterport3D dataset, illustrating one randomly selected panoramic viewpoint per scene.

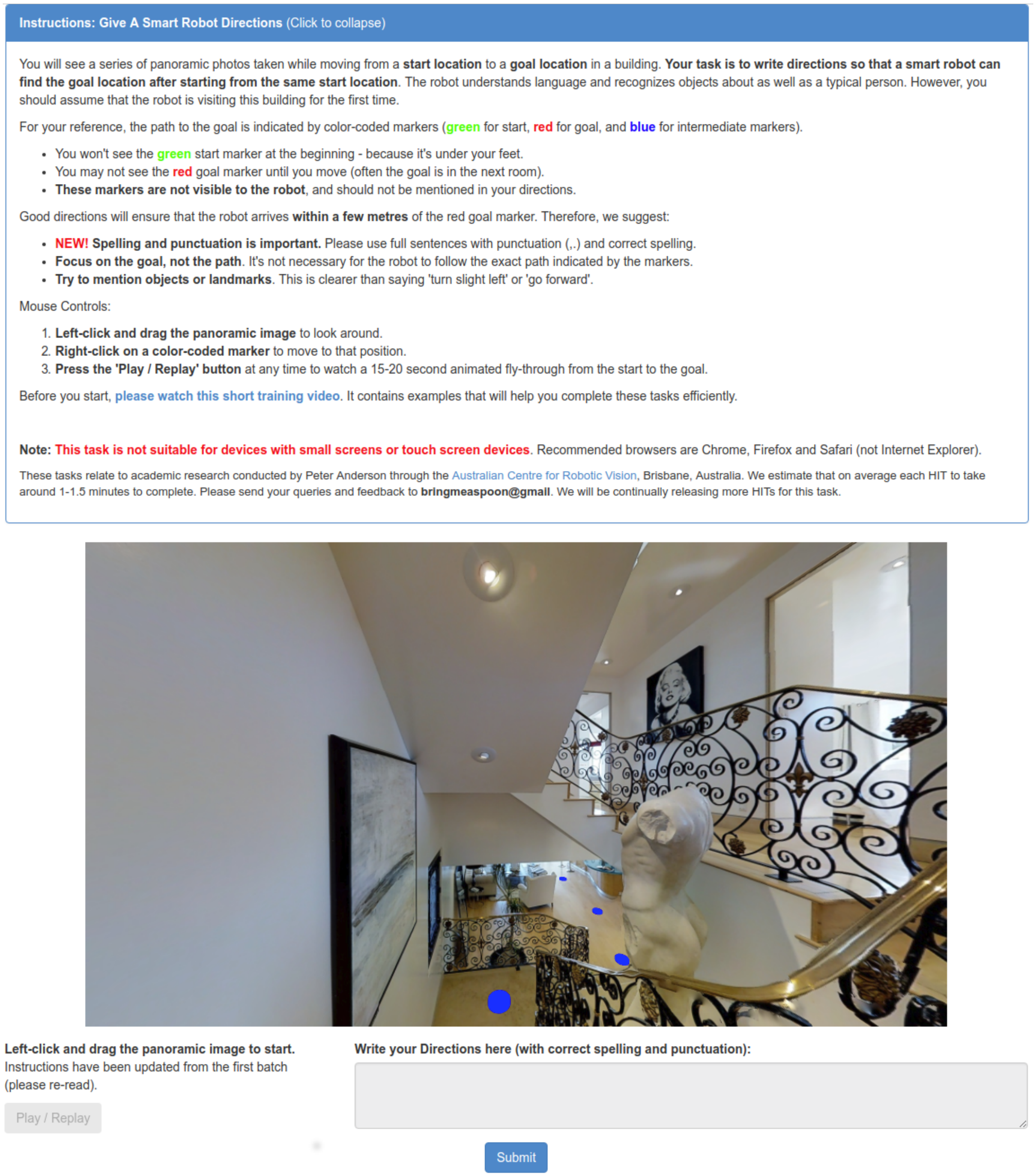

Figure 10. AMT data collection interface for the R2R navigation dataset. Here, blue markers can be seen indicating the trajectory to the goal location. However, in many cases the worker must first look around (pan and tilt) to find the markers. Clicking on a marker moves the camera to that location. Workers can also watch a 'fly-through' of the complete trajectory by clicking the Play / Replay button.